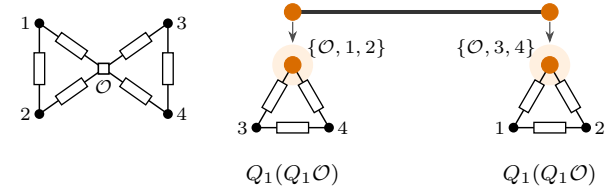
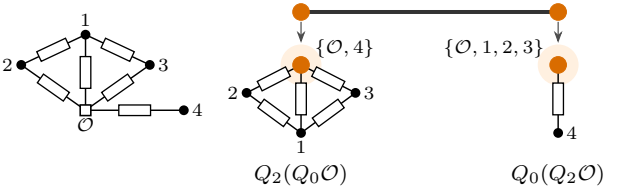
(A) a cut vertex: two ends

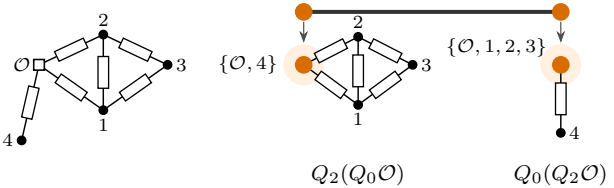
A1



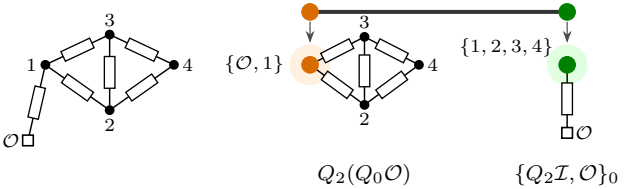
A2



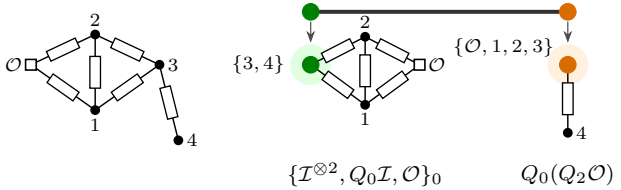
A3



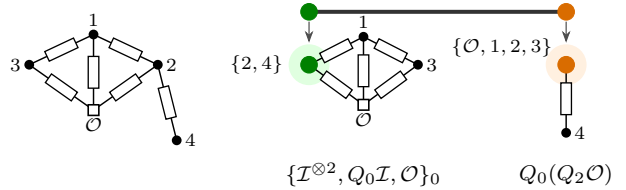
A4



A5

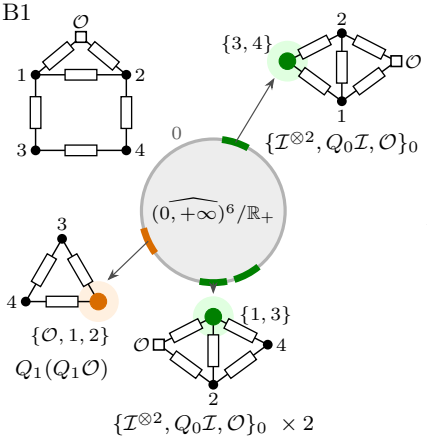


A6

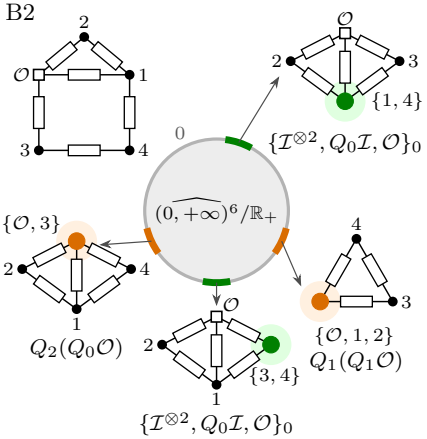


(B) no cut vertex

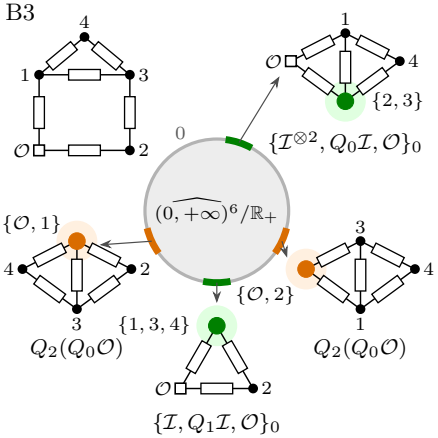
B1



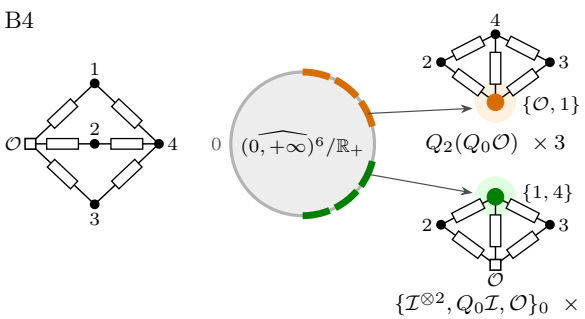
B2



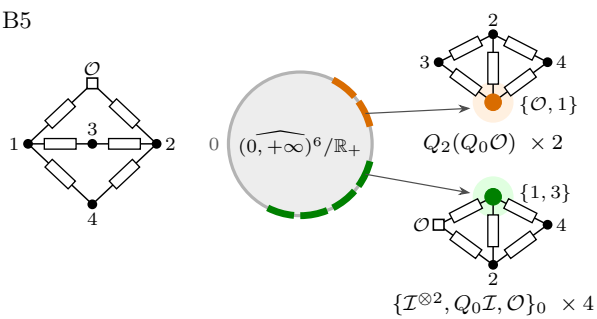
B3



B4



B5



$$\sum_{\text{circuits}} \sum_{\text{faces}} = 0 \implies Q_2Q_0\mathcal{O} + Q_1Q_1\mathcal{O} + Q_0Q_2\mathcal{O} = \tfrac{1}{4}\{\mathcal{I}^{\otimes 2},Q_0\mathcal{I},\mathcal{O}\}_0 + \tfrac{1}{3}\{\mathcal{I},Q_1\mathcal{I},\mathcal{O}\}_0 + \tfrac{1}{4}\{Q_2\mathcal{I},\mathcal{O}\}_0 = 0$$